

Quant Comp Study

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Question

Suppose you have the nonlinear regression:

$$y_i = e^{\beta x_i} + u_i, \quad E[u_i | x_i] = 0$$

where u_i are zero mean IID error terms and IID data (x_i, y_i) for $i = 1, \dots, n$. Describe in detail how you could estimate β .

We start with the linearized model:

$$\ln(y_i) = x_i \beta + v_i$$

where v_i is the error term. Assume that heteroskedasticity has been addressed (e.g., using robust standard errors or Weighted Least Squares), ensuring that OLS is unbiased and efficient.

Step 1: Define the Sum of Squared Residuals (SSR)

The residual is:

$$v_i = \ln(y_i) - x_i \beta$$

The Sum of Squared Residuals (SSR) is:

$$\text{SSR} = \sum_{i=1}^n (\ln(y_i) - x_i \beta)^2$$

Step 2: Expand SSR

Expand the squared term using the distributive property (FOIL):

$$\begin{aligned} & (\ln(y_i) - x_i\beta)(\ln(y_i) - x_i\beta) \\ \text{SSR} &= \sum_{i=1}^n [\ln(y_i)^2 - 2\ln(y_i)x_i\beta + (x_i\beta)^2] \end{aligned}$$

Step 3: Simplify the Expression

Separate the terms in the summation:

$$\text{SSR} = \sum_{i=1}^n \ln(y_i)^2 - 2\beta \sum_{i=1}^n \ln(y_i)x_i + \beta^2 \sum_{i=1}^n x_i^2$$

Step 4: Differentiate with Respect to β

Take the derivative of SSR with respect to β :

$$\frac{\partial \text{SSR}}{\partial \beta} = -2 \sum_{i=1}^n \ln(y_i)x_i + 2\beta \sum_{i=1}^n x_i^2$$

Set the derivative equal to zero to find the minimum:

$$-2 \sum_{i=1}^n \ln(y_i)x_i + 2\beta \sum_{i=1}^n x_i^2 = 0$$

Step 5: Solve for β

Rearrange the equation:

$$2\beta \sum_{i=1}^n x_i^2 = 2 \sum_{i=1}^n \ln(y_i)x_i$$

Divide through by 2:

$$\beta \sum_{i=1}^n x_i^2 = \sum_{i=1}^n \ln(y_i)x_i$$

Solve for β :

$$\beta = \frac{\sum_{i=1}^n \ln(y_i)x_i}{\sum_{i=1}^n x_i^2}$$

Final OLS Estimator

The OLS estimator for the linearized model is:

$$\hat{\beta} = \frac{\sum_{i=1}^n \ln(y_i) x_i}{\sum_{i=1}^n x_i^2}$$

Question

Suppose you have the nonlinear regression:

$$y_i = \alpha(1 + \beta)^{(3x_i - 1)} + u_i, \quad E[u_i | x_i] = 0,$$

where u_i are zero mean IID error terms and (x_i, y_i) are IID data for $i = 1, \dots, n$. Describe in detail how you could estimate α and β .

Solution

We are tasked with estimating α and β from the nonlinear regression model:

$$y_i = \alpha(1 + \beta)^{(3x_i - 1)} + u_i, \quad E[u_i | x_i] = 0,$$

where u_i are zero mean IID error terms and (x_i, y_i) are IID data for $i = 1, \dots, n$.

Step 1: Linearize the Model

To simplify the estimation, we take the natural logarithm of both sides:

$$\ln(y_i) = \ln \left[\alpha(1 + \beta)^{(3x_i - 1)} \right] + u_i.$$

Using the properties of logarithms:

$$\ln(y_i) = \ln(\alpha) + (3x_i - 1) \ln(1 + \beta) + u_i.$$

Let:

$$z_i = \ln(y_i), \quad c = \ln(\alpha), \quad d = \ln(1 + \beta).$$

The model becomes:

$$z_i = c + (3x_i - 1)d + u_i.$$

Thus, the problem is now a linear regression problem where $c = \ln(\alpha)$ and $d = \ln(1 + \beta)$.

Step 2: Define the Residuals and Sum of Squared Residuals (SSR)

Define the residual as:

$$u_i = z_i - (c + (3x_i - 1)d).$$

The sum of squared residuals is:

$$SSR = \sum_{i=1}^n [z_i - (c + (3x_i - 1)d)]^2.$$

Step 3: Expand the SSR Using the Distributive Property (FOIL)

Expanding the SSR:

$$SSR = \sum_{i=1}^n \left[z_i^2 - 2z_i(c + (3x_i - 1)d) + (c + (3x_i - 1)d)^2 \right].$$

Breaking this into terms:

$$SSR = \sum_{i=1}^n z_i^2 - 2c \sum_{i=1}^n z_i - 2d \sum_{i=1}^n z_i(3x_i - 1) + \sum_{i=1}^n [c^2 + 2c(3x_i - 1)d + d^2(3x_i - 1)^2].$$

Step 4: Differentiate SSR with Respect to c and d

To minimize the SSR, differentiate with respect to c and d , and set the derivatives to zero.

Derivative with respect to c :

$$\frac{\partial SSR}{\partial c} = -2 \sum_{i=1}^n z_i + 2c \sum_{i=1}^n 1 + 2d \sum_{i=1}^n (3x_i - 1) = 0.$$

Simplify:

$$-2 \sum_{i=1}^n z_i + 2nc + 2d \sum_{i=1}^n (3x_i - 1) = 0.$$

Derivative with respect to d :

$$\frac{\partial SSR}{\partial d} = -2 \sum_{i=1}^n z_i(3x_i - 1) + 2c \sum_{i=1}^n (3x_i - 1) + 2d \sum_{i=1}^n (3x_i - 1)^2 = 0.$$

Simplify:

$$-2 \sum_{i=1}^n z_i(3x_i - 1) + 2c \sum_{i=1}^n (3x_i - 1) + 2d \sum_{i=1}^n (3x_i - 1)^2 = 0.$$

Step 5: Solve the System of Equations for c and d

We now have two equations:

$$\sum_{i=1}^n z_i = nc + d \sum_{i=1}^n (3x_i - 1),$$

$$\sum_{i=1}^n z_i(3x_i - 1) = c \sum_{i=1}^n (3x_i - 1) + d \sum_{i=1}^n (3x_i - 1)^2.$$

These can be solved simultaneously to obtain c and d :

$$c = \ln(\alpha), \quad d = \ln(1 + \beta).$$

Step 6: Back-transform to Estimate α and β

From $c = \ln(\alpha)$ and $d = \ln(1 + \beta)$, solve for α and β :

$$\alpha = e^c, \quad \beta = e^d - 1.$$

Thus, the estimates of α and β are obtained.

Question

Suppose you have the nonlinear regression:

$$y_i = \alpha(2\beta - 7)^{(5x_i - 4)} + u_i, \quad E[u_i|x_i] = 0,$$

where u_i are zero mean IID error terms and (x_i, y_i) are IID data for $i = 1, \dots, n$. Describe in detail how you could estimate α and β .

Solution

We are tasked with estimating α and β from the nonlinear regression model:

$$y_i = \alpha(2\beta - 7)^{(5x_i - 4)} + u_i, \quad E[u_i|x_i] = 0,$$

where u_i are zero mean IID error terms and (x_i, y_i) are IID data for $i = 1, \dots, n$.

Step 1: Linearize the Model

To simplify the estimation, we take the natural logarithm of both sides:

$$\ln(y_i) = \ln \left[\alpha(2\beta - 7)^{(5x_i - 4)} \right] + u_i.$$

Using the properties of logarithms:

$$\ln(y_i) = \ln(\alpha) + (5x_i - 4) \ln(2\beta - 7) + u_i.$$

Let:

$$z_i = \ln(y_i), \quad c = \ln(\alpha), \quad d = \ln(2\beta - 7).$$

The model becomes:

$$z_i = c + (5x_i - 4)d + u_i.$$

Thus, the problem is now a linear regression problem where $c = \ln(\alpha)$ and $d = \ln(2\beta - 7)$.

Step 2: Define the Residuals and Sum of Squared Residuals (SSR)

Define the residual as:

$$u_i = z_i - (c + (5x_i - 4)d).$$

The sum of squared residuals is:

$$SSR = \sum_{i=1}^n [z_i - (c + (5x_i - 4)d)]^2.$$

Step 3: Expand the SSR Using the Distributive Property (FOIL)

Expanding the SSR:

$$SSR = \sum_{i=1}^n \left[z_i^2 - 2z_i(c + (5x_i - 4)d) + (c + (5x_i - 4)d)^2 \right].$$

Breaking this into terms:

$$SSR = \sum_{i=1}^n z_i^2 - 2c \sum_{i=1}^n z_i - 2d \sum_{i=1}^n z_i(5x_i - 4) + \sum_{i=1}^n [c^2 + 2c(5x_i - 4)d + d^2(5x_i - 4)^2].$$

Step 4: Differentiate SSR with Respect to c and d

To minimize the SSR, differentiate with respect to c and d , and set the derivatives to zero.

Derivative with respect to c :

$$\frac{\partial SSR}{\partial c} = -2 \sum_{i=1}^n z_i + 2c \sum_{i=1}^n 1 + 2d \sum_{i=1}^n (5x_i - 4) = 0.$$

Simplify:

$$-2 \sum_{i=1}^n z_i + 2nc + 2d \sum_{i=1}^n (5x_i - 4) = 0.$$

Derivative with respect to d :

$$\frac{\partial SSR}{\partial d} = -2 \sum_{i=1}^n z_i (5x_i - 4) + 2c \sum_{i=1}^n (5x_i - 4) + 2d \sum_{i=1}^n (5x_i - 4)^2 = 0.$$

Simplify:

$$-2 \sum_{i=1}^n z_i (5x_i - 4) + 2c \sum_{i=1}^n (5x_i - 4) + 2d \sum_{i=1}^n (5x_i - 4)^2 = 0.$$

Step 5: Solve the System of Equations for c and d

We now have two equations:

$$\begin{aligned} \sum_{i=1}^n z_i &= nc + d \sum_{i=1}^n (5x_i - 4), \\ \sum_{i=1}^n z_i (5x_i - 4) &= c \sum_{i=1}^n (5x_i - 4) + d \sum_{i=1}^n (5x_i - 4)^2. \end{aligned}$$

These can be solved simultaneously to obtain c and d :

$$c = \ln(\alpha), \quad d = \ln(2\beta - 7).$$

Step 6: Back-transform to Estimate α and β

From $c = \ln(\alpha)$ and $d = \ln(2\beta - 7)$, solve for α and β :

$$\alpha = e^c, \quad \beta = \frac{e^d + 7}{2}.$$

Thus, the estimates of α and β are obtained.

Question

Consider the following data generating process:

$$Y_t = \varepsilon_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2}$$

where ε_t is a white noise process. Assuming $0 < \phi_1 < 1$ and $0 < \phi_2 < 1$, answer the following:

- (a) Derive the expected value of this process.
- (b) Derive the variance of this process.
- (c) Derive the covariance of this process for $j = 1, 2, 3$.
- (d) Derive the autocorrelation function of this process for $j = 1, 2, 3$.

Solution

Part (a): Derive the Expected Value of the Process

The data generating process is:

$$Y_t = \phi_3 Y_{t-3} + \varepsilon_t, \quad \text{where } 0 < \phi_3 < 1$$

Taking the expectation of both sides:

$$E[Y_t] = E[\phi_3 Y_{t-3} + \varepsilon_t]$$

$$E[Y_t] = \phi_3 E[Y_{t-3}] + E[\varepsilon_t]$$

Since ε_t is white noise, $E[\varepsilon_t] = 0$:

$$E[Y_t] = \phi_3 E[Y_{t-3}]$$

Using the stationarity assumption ($E[Y_t] = E[Y_{t-3}]$):

$$E[Y_t] = \phi_3 E[Y_t]$$

Rearranging:

$$E[Y_t](1 - \phi_3) = 0$$

Given the assumption $0 < \phi_3 < 1$, this implies:

$$E[Y_t] = 0$$

Part (b): Derive the Variance of the Process

The data generating process is:

$$Y_t = \phi_3 Y_{t-3} + \varepsilon_t$$

Taking the variance of both sides:

$$\text{Var}(Y_t) = \text{Var}(\phi_3 Y_{t-3} + \varepsilon_t)$$

Using the properties of variance:

$$\text{Var}(Y_t) = \phi_3^2 \text{Var}(Y_{t-3}) + \text{Var}(\varepsilon_t)$$

Since the process is stationary ($\text{Var}(Y_t) = \text{Var}(Y_{t-3}) = \text{Var}(Y)$), this simplifies to:

$$\text{Var}(Y) = \phi_3^2 \text{Var}(Y) + \sigma^2$$

Rearranging:

$$\text{Var}(Y)(1 - \phi_3^2) = \sigma^2$$

Solving for $\text{Var}(Y)$:

$$\text{Var}(Y) = \frac{\sigma^2}{1 - \phi_3^2}$$

Thus, the variance of the process is:

$$\text{Var}(Y) = \frac{\sigma^2}{1 - \phi_3^2}$$

Part (c): Derive the Covariance of the Process for $j = 1, 2, 3$

The data generating process is:

$$Y_t = \phi_3 Y_{t-3} + \varepsilon_t$$

Using the definition of covariance:

$$\text{Cov}(Y_t, Y_{t-j}) = E[Y_t Y_{t-j}] - E[Y_t]E[Y_{t-j}]$$

Since $E[Y_t] = 0$ (from part (a)), this simplifies to:

$$\text{Cov}(Y_t, Y_{t-j}) = E[Y_t Y_{t-j}]$$

Substituting $Y_t = \phi_3 Y_{t-3} + \varepsilon_t$:

$$E[Y_t Y_{t-j}] = E[(\phi_3 Y_{t-3} + \varepsilon_t) Y_{t-j}]$$

Expanding the terms:

$$E[Y_t Y_{t-j}] = \phi_3 E[Y_{t-3} Y_{t-j}] + E[\varepsilon_t Y_{t-j}]$$

Since ε_t is white noise and uncorrelated with Y_{t-j} for $j > 0$:

$$E[\varepsilon_t Y_{t-j}] = 0$$

Thus:

$$E[Y_t Y_{t-j}] = \phi_3 E[Y_{t-3} Y_{t-j}]$$

For $j = 1, 2, 3$, this recursive relationship continues until we reach terms proportional to ϕ_3 , allowing us to calculate $\text{Cov}(Y_t, Y_{t-j})$ explicitly.

Part (d): Derive the Autocorrelation Function for $j = 1, 2, 3$

The autocorrelation function (ACF) is defined as:

$$\rho_j = \frac{\text{Cov}(Y_t, Y_{t-j})}{\text{Var}(Y_t)}$$

From part (b), the variance of Y_t is:

$$\text{Var}(Y_t) = \frac{\sigma^2}{1 - \phi_3^2}$$

Thus, the autocorrelation for lag j is:

$$\rho_j = \frac{\text{Cov}(Y_t, Y_{t-j})}{\frac{\sigma^2}{1 - \phi_3^2}}$$

Simplifying further:

$$\rho_j = (1 - \phi_3^2) \cdot \frac{\text{Cov}(Y_t, Y_{t-j})}{\sigma^2}$$

Using the recursive expression for $\text{Cov}(Y_t, Y_{t-j})$ from part (c), we compute ρ_j for $j = 1, 2, 3$ as a function of ϕ_3 and σ^2 .

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For $j = 1$:

$$\rho_1 = \phi_3$$

For $j = 2$:

$$\rho_2 = \phi_3^2$$

For $j = 3$:

$$\rho_3 = \phi_3^3$$

Thus, the autocorrelation function is:

$$\rho_j = \phi_3^j \quad \text{for } j = 1, 2, 3$$

c)

$$\text{Cov}(Y_t, Y_{t-j})$$

$$- j = 3$$

$$\rightarrow \text{Cov}(Y_t, Y_{t-3}) \text{ given } Y_t = \rho_3 Y_{t-3} + \varepsilon_t$$

$$\rightarrow \text{Cov}(Y_t, Y_{t-3}) = \text{Cov}(\rho_3 Y_{t-3} + \varepsilon_t, Y_{t-3})$$

$$\rightarrow \text{Cov}(Y_{t-3}, \varepsilon_t) = 0$$

$$\rightarrow \text{Cov}(Y_t, Y_{t-3}) = \rho_3 \text{Var}(Y_{t-3})$$

$$\rightarrow \text{Cov}(Y_t, Y_{t-3}) = \rho_3 \text{Var}(Y_{t-3}) = \boxed{\rho_3 \sigma^2}$$

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$$- j = 2$$

$$\text{Cov}(Y_t, Y_{t-2}) \neq \text{Cov}(\rho_3 Y_{t-3} + \varepsilon_t, Y_{t-2})$$

$$\boxed{\text{Cov}(Y_t, Y_{t-2}) = 0}$$

$$- j = 1$$

$$\text{Cov}(Y_t, Y_{t-1}) \neq \text{Cov}(\rho_3 Y_{t-3} + \varepsilon_t, Y_{t-1})$$

$$\boxed{\text{Cov}(Y_t, Y_{t-1}) = 0}$$

Figure 1: part c

d)

auto correlation $\rightarrow \rho(j) = \frac{\text{cov}(Y_t, Y_{t-j})}{\text{var}(Y)}$

- $j = 1$

$$\rho(1) = \frac{\text{cov}(Y_t, Y_{t-1})}{\text{var}(Y)}$$

where Y_t does not depend on Y_{t-1} .

$$\text{Thus } \rho(1) = 0$$

- $j = 2$

$$\rho(2) = \frac{\text{cov}(Y_t, Y_{t-2})}{\text{var}(Y)}$$

where Y_t does not depend on Y_{t-2} .

$$\text{Thus } \rho(2) = 0$$

- $j = 3$

$$\rho(3) = \frac{\text{cov}(Y_t, Y_{t-3})}{\text{var}(Y)} = \frac{\phi_3 \text{var}(Y)}{\text{var}(Y)}$$

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$$\rho(3) = \frac{\text{cov}(Y_t, Y_{t-3})}{\text{var}(Y)} = \frac{\phi_3 \text{var}(Y)}{\text{var}(Y)}$$

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$$= \frac{\phi_3 \sigma^2}{\sigma^2}$$

$$\rho(3) = \phi_3$$

Question

In 1985, neither Florida nor Georgia had laws banning open alcohol containers in vehicle passenger compartments. By 1990, Florida had passed such a law, but Georgia had not. Suppose you can collect random samples of the driving-age population in both states, for 1985 and 1990. Let *arrest* be a binary variable equal to unity if a person was arrested for drunk driving during the year.

[label=()]Using a difference-in-differences (DiD) identification strategy, write down a linear probability model (LPM) that allows you to test whether the open container law reduced the probability of being arrested for drunk driving. Which coefficient in your model measures the effect of the law? What is the interpretation of the coefficient of interest in your LPM model? How will the interpretation be different if you use a Logit model instead of the LPM? What are the advantages and disadvantages of the LPM when compared to other models with limited dependent variables?

A DiD measures the causal effect of a change in an intervention or policy. An LPM model is simple regression measuring the probability of a binary outcome. This question is investigated by comparing arrests for drunk driving in Florida and Georgia in 1985 and 1990. A linear probability model that uses a difference-in-differences (DiD) identification strategy to test whether the open container law reduces drunk driving arrests would look like:

$$Y_{ist} = \alpha + \gamma FL_s + \lambda d_t + \delta (FL_s \cdot d_t) + \varepsilon_{ist}$$

Where,

$$\alpha = E[Y_{ist} | s = GA, t = 1985] = \gamma_{GA} + \lambda_{1985}$$

$$\begin{aligned}\gamma &= E[Y_{ist}|s = FL, t = 1985] - E[Y_{ist}|s = GA, t = 1985] = \gamma_{FL} - \gamma_{GA} \\ \lambda &= E[Y_{ist}|s = GA, t = 1990] - E[Y_{ist}|s = GA, t = 1985] = \lambda_{1990} - \lambda_{1985} \\ \delta &= E[Y_{ist}|s = FL, t = 1990] - E[Y_{ist}|s = FL, t = 1985] - E[Y_{ist}|s = GA, t = 1990] + E[Y_{ist}|s = GA, t = 1985] \\ &= \lambda_{1990} - \lambda_{1985}\end{aligned}$$

The coefficient δ measures the effect of the law.

The linear probability model (LPM) uses a binary dependent variable, which allows the coefficients to describe the effect of explanatory variables on the probability that $y = 1$. The LPM enables easy estimation and interpretation. The estimated effects and predictions are often reasonably good in practice. The disadvantages however of the LPM is that predicted probabilities may be larger than one or smaller than zero and the marginal probability effects are sometimes logically impossible. Moreover, for binary dependent variables, predictions sometimes lie outside of the unit interval and the partial effects of explanatory variables are constant.

Question

What is the difference between Sharp and Fuzzy RD strategy? Write down equations showing that a Fuzzy RD model leads to a simple 2SLS estimation strategy. Explain the first stage equation, the reduced form equation, the instrument, and the instrumented variables in your model.

Sharp vs Fuzzy Regression Discontinuity Design

Sharp RDD: B_2 represents the causal effect at the cutoff (threshold) on the outcome Y .

Fuzzy RDD: Y represents the causal effect of the treatment on the treated group, adjusted for the fact that not all individuals receive the treatment based on the threshold.

Sharp regression discontinuity research design is used when the treatment status is a deterministic and discontinuous function of a covariate, x_i , and x_0 is a known threshold or cutoff. This assignment mechanism is deterministic because once we know x_i , we know D_i .

$$D_i = \begin{cases} 1 & \text{if } x_i \geq x_0, \\ 0 & \text{if } x_i < x_0 \end{cases}$$

Treatment is discontinuous because treatment remains unchanged until $x_i = x_0$. The study on graduate school attendance rates of students with PSAT scores just above and below the National Merit Award thresholds deploys a Sharp RD.

Fuzzy Regression Discontinuity Design

A Fuzzy RD, however, exploits discontinuities in the probability or expected value of treatment conditional on a covariate. In this way, the discontinuity becomes an instrumental variable for treatment states instead of deterministically switching on or off, as in a Sharp RD.

$$P(D_i = 1|x_i) = \begin{cases} g_1(x_i) & \text{if } x_i \geq x_0, \\ g_0(x_i) & \text{if } x_i < x_0, \end{cases} \quad \text{where } g_1(x_0) \neq g_0(x_0)$$

For instance, in Angrist and Lavy's 1999 study, they exploit the discontinuities in Maimonides Rule, an arbitrary mathematics principle. This Fuzzy RD research design estimates the effects of class size on children's test scores in Israel by breaking down to a 2SLS estimation strategy.

Fuzzy RD and 2SLS

The Fuzzy RD model can be written as a 2SLS system:

First Stage:

$$D_i = \pi_0 + \pi_1 \mathbb{I}(x_i \geq x_0) + \pi_2 x_i + \pi_3 Z_i + \nu_i,$$

where D_i is the treatment, and $\mathbb{I}(x_i \geq x_0)$ is the instrument.

Second Stage:

$$Y_i = \beta_0 + \beta_1 \hat{D}_i + \beta_2 x_i + \beta_3 Z_i + \varepsilon_i,$$

where $\hat{D}_i$ is the predicted treatment from the first stage, and β_1 measures the causal effect of D_i on Y_i .

The exclusion restriction requires $\mathbb{I}(x_i \geq x_0)$ to affect Y_i only through D_i .

Question

Prove that R^2 associated with a restricted least squares estimator is never larger than that associated with the unrestricted least squares estimator.

Proof with Mathematical Steps and Example

Step 1: Definition of R^2

The coefficient of determination, R^2 , is defined as:

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}},$$

where:

- SSR is the sum of squared residuals: $\text{SSR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$,
- SST is the total sum of squares: $\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2$.

The R^2 measures the proportion of total variation in the dependent variable (y_i) that is explained by the model. The smaller the SSR, the larger the R^2 .

Step 2: Effect of Restrictions on SSR

Unrestricted Model: The unrestricted least squares model minimizes SSR without any constraints.

Restricted Model: The restricted model minimizes SSR under additional constraints (e.g., some coefficients are set to zero).

Since the restricted model is solving a constrained optimization problem, it cannot achieve a lower SSR than the unrestricted model. Therefore:

$$\text{SSR}_{\text{restricted}} \geq \text{SSR}_{\text{unrestricted}}.$$

Step 3: Implications for R^2

Using the formula for R^2 :

$$R_{\text{restricted}}^2 = 1 - \frac{\text{SSR}_{\text{restricted}}}{\text{SST}},$$

$$R_{\text{unrestricted}}^2 = 1 - \frac{\text{SSR}_{\text{unrestricted}}}{\text{SST}}.$$

Since $\text{SSR}_{\text{restricted}} \geq \text{SSR}_{\text{unrestricted}}$, it follows that:

$$\frac{\text{SSR}_{\text{restricted}}}{\text{SST}} \geq \frac{\text{SSR}_{\text{unrestricted}}}{\text{SST}},$$

and therefore:

$$R_{\text{restricted}}^2 \leq R_{\text{unrestricted}}^2.$$

Step 4: Practical Example

To illustrate, consider the following data:

Observation (i)	y_i	x_i
1	10	2
2	15	4
3	20	6
4	25	8

Unrestricted Model: Let the unrestricted model be:

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

The OLS estimates of β_0 and β_1 are derived using the least squares normal equations:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

For this dataset:

$$\bar{x} = \frac{2 + 4 + 6 + 8}{4} = 5, \quad \bar{y} = \frac{10 + 15 + 20 + 25}{4} = 17.5.$$

Calculate $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - 5)(y_i - 17.5)}{\sum_{i=1}^n (x_i - 5)^2}.$$

First, compute the terms:

$$\sum_{i=1}^n (x_i - 5)(y_i - 17.5) = (2-5)(10-17.5) + (4-5)(15-17.5) + (6-5)(20-17.5) + (8-5)(25-17.5) = 35.$$

$$\sum_{i=1}^n (x_i - 5)^2 = (2 - 5)^2 + (4 - 5)^2 + (6 - 5)^2 + (8 - 5)^2 = 20.$$

Thus:

$$\hat{\beta}_1 = \frac{35}{20} = 1.75.$$

Calculate $\hat{\beta}_0$:

$$\hat{\beta}_0 = 17.5 - 1.75(5) = 8.75.$$

The unrestricted model is:

$$\hat{y}_i = 8.75 + 1.75x_i.$$

Restricted Model: Suppose the restriction is $\beta_1 = 0$, i.e., the restricted model becomes:

$$y_i = \beta_0 + u_i.$$

The OLS estimate for β_0 in the restricted model is simply the mean of y_i , since the model is just a horizontal line:

$$\hat{\beta}_0 = \bar{y} = 17.5.$$

Comparison of SSR:

For the unrestricted model:

$$SSR_{\text{unrestricted}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 0,$$

since the line perfectly fits the data.

For the restricted model:

$$SSR_{\text{restricted}} = \sum_{i=1}^n (y_i - 17.5)^2 = (10 - 17.5)^2 + (15 - 17.5)^2 + (20 - 17.5)^2 + (25 - 17.5)^2 = 125.$$

Comparison of R^2 :

For the unrestricted model:

$$R^2_{\text{unrestricted}} = 1 - \frac{SSR_{\text{unrestricted}}}{SST} = 1 - \frac{0}{125} = 1.$$

For the restricted model:

$$R^2_{\text{restricted}} = 1 - \frac{SSR_{\text{restricted}}}{SST} = 1 - \frac{125}{125} = 0.$$

Thus:

$$R^2_{\text{restricted}} < R^2_{\text{unrestricted}},$$

which confirms that R^2 for the restricted model is never larger than that of the unrestricted model.

Conceptual Explanation

The restricted least squares model includes fewer explanatory variables compared to the unrestricted model. Consequently, the restricted model has fewer degrees of freedom to explain the variation in the dependent variable Y . This results in a higher sum of squared residuals (SSR) for the restricted model, as it cannot fit the data as closely as the unrestricted model.

$$SSR_{\text{restricted}} \geq SSR_{\text{unrestricted}}$$

The coefficient of determination, R^2 , is defined as:

$$R^2 = 1 - \frac{SSR}{SST}$$

Since the restricted model has a larger SSR, the proportion of variation in Y explained by the restricted model is smaller than that explained by the unrestricted model. Mathematically:

$$R^2_{\text{restricted}} \leq R^2_{\text{unrestricted}}$$

In summary: - The **restricted model** explains less variation in Y because it has fewer variables. - The **unrestricted model** explains more variation in Y because it includes additional variables, allowing for a better fit to the data.

Thus, the R^2 of the restricted model will never exceed the R^2 of the unrestricted model.

Question

Suppose you have a panel data on different countries' institutional quality and GDP per capita. You want to measure the association between institutional quality and country-level economic performance. Write down an IV regression model (write down both the first stage and the reduced form equations) that includes your suggested control variables. What is the economic interpretation of your coefficient of interest? What is the exclusion restriction for your IV model?

Solution

We are tasked with constructing an IV regression model using institutional quality and GDP per capita, as well as addressing the interpretation of the coefficient of interest and the exclusion restriction.

Step 1: First Stage Equation

In the first stage, we use an instrumental variable (Z) to explain the variation in institutional quality (X). The first-stage regression model is written as:

$$X_{it} = \gamma_0 + \gamma_1 Z_{it} + \gamma_2 \text{Controls}_{it} + \eta_{it},$$

where:

- X_{it} : Institutional quality for country i at time t ,
- Z_{it} : Instrumental variable (e.g., share of remote work),
- Controls_{it} : Control variables such as family size, education, geography, and average pay,
- η_{it} : Error term.

The coefficient γ_1 captures how changes in the instrument (Z_{it}) affect institutional quality (X_{it}). The economic interpretation of γ_1 is that an increase in the share of remote work positively or negatively impacts institutional quality.

Step 2: Second Stage Equation (Structural Equation)

The second stage uses the predicted values of institutional quality ($\hat{X}_{it}$) from the first stage to explain GDP per capita (Y_{it}):

$$Y_{it} = \beta_0 + \beta_1 \hat{X}_{it} + \beta_2 \text{Controls}_{it} + \varepsilon_{it},$$

where:

- Y_{it} : GDP per capita for country i at time t ,
- $\hat{X}_{it}$: Predicted institutional quality from the first stage,
- Controls_{it} : Same control variables as in the first stage,
- ε_{it} : Error term.

The coefficient β_1 measures the causal effect of institutional quality on GDP per capita. Its economic interpretation is that an improvement in institutional quality leads to a positive or negative change in GDP per capita, depending on the sign of β_1 .

Step 3: Reduced Form Equation

The reduced form regression examines the direct relationship between the instrument (Z_{it}) and the outcome variable (Y_{it}):

$$Y_{it} = \delta_0 + \delta_1 Z_{it} + \delta_2 \text{Controls}_{it} + \nu_{it}.$$

Here, δ_1 measures the total effect of the instrument (Z_{it}) on GDP per capita (Y_{it}).

Step 4: Exclusion Restriction

The exclusion restriction ensures that the instrument (Z_{it}) affects the outcome (Y_{it}) only through the endogenous variable (X_{it}), not directly. Mathematically:

$$E[\varepsilon_{it}|Z_{it}] = 0.$$

This restriction implies:

- The instrument (Z_{it}) must be uncorrelated with the error term (ε_{it}) in the second stage.
- The instrument (Z_{it}) should not have a direct effect on GDP per capita (Y_{it}) except through institutional quality (X_{it}).

Step 5: Economic Interpretation of the Coefficients

First Stage: The coefficient γ_1 represents the effect of the instrument (Z_{it}) on institutional quality (X_{it}). For example, an increase in the share of remote work may improve institutional quality by allowing more flexible working arrangements.

Second Stage: The coefficient β_1 represents the causal effect of institutional quality (X_{it}) on GDP per capita (Y_{it}). For instance, an improvement in institutional quality may lead to better economic performance, as reflected in higher GDP per capita.

Step 6: Instrument Validity and Exclusion Restriction

For the IV model to be valid, the following conditions must hold:

1. **Relevance:** The instrument (Z_{it}) must be correlated with the endogenous variable (X_{it}), i.e., $\text{Cov}(Z_{it}, X_{it}) \neq 0$.
2. **Exclusion Restriction:** The instrument (Z_{it}) must not be correlated with the error term (ε_{it}) in the second stage, i.e., $\text{Cov}(Z_{it}, \varepsilon_{it}) = 0$.

For example, if remote work (Z_{it}) is used as an instrument, we argue that it affects GDP per capita only through its impact on institutional quality, satisfying the exclusion restriction.

Question

Consider the regression model:

$$Y_i = \beta X_i + u_i, \quad i = 1, \dots, n,$$

where u_i and X_i satisfy the standard assumptions. Let:

$$\hat{\beta} = \frac{\bar{Y}}{\bar{X}}, \quad \text{where} \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \quad \text{and} \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Answer the following:

- (a) Show that $\hat{\beta}$ is a linear function of $Y_1, Y_2, \dots, Y_n$.
 - (b) Show that $\hat{\beta}$ is conditionally unbiased.
 - (c) Derive the variance of $\hat{\beta}$.
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Solution

(a) Show that $\hat{\beta}$ is a linear function of $Y_1, Y_2, \dots, Y_n$

The estimator $\hat{\beta}$ can be written as:

$$\hat{\beta} = \frac{\bar{Y}}{\bar{X}} = \frac{\frac{1}{n} \sum_{i=1}^n Y_i}{\frac{1}{n} \sum_{i=1}^n X_i}.$$

Expanding $\bar{Y}$:

$$\hat{\beta} = \frac{\sum_{i=1}^n Y_i}{\sum_{i=1}^n X_i}.$$

Since $\hat{\beta}$ depends linearly on the Y_i values through $\sum_{i=1}^n Y_i$, it is a linear function of $Y_1, Y_2, \dots, Y_n$.

(b) Show that $\hat{\beta}$ is conditionally unbiased

The expectation of $\hat{\beta}$ is:

$$E[\hat{\beta}] = E\left[\frac{\bar{Y}}{\bar{X}}\right].$$

Using the regression model $Y_i = \beta X_i + u_i$:

$$\bar{Y} = \beta \bar{X} + \bar{u}, \quad \text{where} \quad \bar{u} = \frac{1}{n} \sum_{i=1}^n u_i.$$

Substitute $\bar{Y}$:

$$\hat{\beta} = \frac{\beta \bar{X} + \bar{u}}{\bar{X}} = \beta + \frac{\bar{u}}{\bar{X}}.$$

Taking the expectation, and noting that $E[\bar{u}] = 0$ under standard assumptions:

$$E[\hat{\beta}] = \beta + E\left[\frac{\bar{u}}{\bar{X}}\right] = \beta.$$

Thus, $\hat{\beta}$ is conditionally unbiased.

(c) Derive the variance of $\hat{\beta}$

From $\hat{\beta} = \beta + \frac{\bar{u}}{\bar{X}}$, the variance is:

$$\text{Var}(\hat{\beta}) = \text{Var}\left(\frac{\bar{u}}{\bar{X}}\right).$$

Since $\bar{u} = \frac{1}{n} \sum_{i=1}^n u_i$, and assuming $\text{Var}(u_i) = \sigma_u^2$:

$$\text{Var}(\bar{u}) = \frac{\sigma_u^2}{n}.$$

Thus:

$$\text{Var}(\hat{\beta}) = \frac{\text{Var}(\bar{u})}{\bar{X}^2} = \frac{\sigma_u^2}{n\bar{X}^2}.$$

Question

Suppose you collect data from a survey on wages, education, experience, and gender. In addition, you ask for information about marijuana usage. The original question is:

“On how many separate occasions last month did you smoke marijuana?”

Answer the following:

- (a) Write an equation that would allow you to estimate the effects of marijuana usage on wage, while controlling for other factors. You should be able to make statements such as, “Smoking marijuana five more times per month is estimated to change wage by $x\%$.”
 - (b) Write a model that would allow you to test whether drug usage has different effects on wages for men and women. How would you test that there are no differences in the effects of drug usage for men and women?
 - (c) Suppose you think it is better to measure marijuana usage by putting people into one of four categories: nonuser, light user (1 to 5 times per month), moderate user (6 to 10 times per month), and heavy user (more than 10 times per month). Now, write a model that allows you to estimate the effects of marijuana usage on wage.
 - (d) Using the model in part (c), explain in detail how to test the null hypothesis that marijuana usage has no effect on wage. Be very specific and include a careful listing of degrees of freedom.
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Answer

(a) Estimating the effects of marijuana usage on wage

We can use a simple Ordinary Least Squares (OLS) regression model:

$$w_i = \alpha + \beta_1 m_i + \Delta Z_i + \varepsilon_i,$$

where:

- w_i : Wage for individual i ,
- m_i : Marijuana usage (number of times per month) for individual i ,
- Z_i : Vector of control variables (e.g., education, experience, gender),
- ε_i : Error term.

The coefficient β_1 represents the effect of marijuana usage on wage. For example, if $\beta_1 = -0.02$, smoking marijuana five more times per month would reduce wage by $5 \times 0.02 = 0.10$ or 10%.

(b) Testing differences in effects for men and women

To test for different effects of marijuana usage on wages by gender, we can use an interaction term in a difference-in-differences framework:

$$w_{is} = \alpha + \beta_1 m_i + \lambda s_i + \delta(m_i \cdot s_i) + \varepsilon_{is},$$

where:

- s_i : Dummy variable for gender (1 if female, 0 if male),
- $m_i \cdot s_i$: Interaction term between marijuana usage and gender.

The coefficient δ captures the differential effect of marijuana usage on wages for women compared to men. To test for no differences, we use the null hypothesis:

$$H_0 : \delta = 0 \quad \text{vs.} \quad H_a : \delta \neq 0.$$

(c) Estimating effects with categorical marijuana usage

To incorporate categorical marijuana usage, we create dummy variables for each category:

$$w_i = \alpha + \beta_1 \text{Light}_i + \beta_2 \text{Moderate}_i + \beta_3 \text{Heavy}_i + \Delta Z_i + \varepsilon_i,$$

where:

- Light_i : Dummy variable for light users (1–5 times per month),
- Moderate_i : Dummy variable for moderate users (6–10 times per month),
- Heavy_i : Dummy variable for heavy users (more than 10 times per month),
- Nonusers are the reference category.

The coefficients β_1 , β_2 , and β_3 measure the effects of light, moderate, and heavy usage relative to nonusers.

(d) Testing the null hypothesis

Using the model in part (c), we test the null hypothesis:

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0 \quad \text{vs.} \quad H_a : \text{At least one } \beta_k \neq 0.$$

We perform an F-test. The test statistic is:

$$F = \frac{\frac{\text{SSR}_{\text{restricted}} - \text{SSR}_{\text{unrestricted}}}{q}}{\frac{\text{SSR}_{\text{unrestricted}}}{n-k}},$$

where:

- $SSR_{\text{restricted}}$: Sum of squared residuals for the restricted model (H_0 is true),
- $SSR_{\text{unrestricted}}$: Sum of squared residuals for the unrestricted model,
- q : Number of restrictions (in this case, $q = 3$),
- n : Number of observations,
- k : Number of parameters in the unrestricted model.

Degrees of freedom for the F-test:

$$df_1 = q, \quad df_2 = n - k.$$

For example, if $n = 300$ and $k = 5$ (including intercept and controls), then:

$$df_1 = 3, \quad df_2 = 300 - 5 = 295.$$

If the F-statistic exceeds the critical value, we reject H_0 , concluding that marijuana usage affects wages.

Question

Regarding coefficients' standard errors:

- When should you adjust standard errors for clustering? Your answer should include a panel regression model that requires adjusting for clusters.
- What are robust standard errors? If we adjust robust standard errors for clustering effects, what will happen to the regression coefficients' size and significance?

Answer

(a) When to adjust standard errors for clustering

Standard errors should be adjusted for clustering when working with panel data or grouped data where observations within clusters may not be independent. Adjusting for clustering accounts for within-cluster correlation, ensuring unbiased and consistent standard errors. A typical case involves panel data with country-level fixed effects or time-invariant factors, such as a person's birth date, sex, or eye color.

A panel regression model with clustering can be written as:

$$y_{ct} = \alpha + \beta_1 x_{ct} + \beta_2 C_c + \beta_3 T_t + \Delta Z_{ct} + \varepsilon_{ct},$$

where:

- y_{ct} : GDP per capita for country c at time t ,
- x_{ct} : Institutional quality for country c at time t ,
- C_c : Country fixed effects (e.g., $C_c = 1$ if observation belongs to country c),
- T_t : Time fixed effects (e.g., $T_t = 1$ if observation corresponds to year t),
- Z_{ct} : Matrix of control variables (e.g., population, trade openness),
- ε_{ct} : Error term.

Clustering standard errors by country ensures that within-country correlations across time are accounted for, avoiding underestimated standard errors.

(b) Robust standard errors and clustering effects

Robust standard errors adjust for heteroskedasticity, ensuring consistency even if the error variance is not constant. When robust standard errors are adjusted for clustering effects:

- The standard errors may increase or decrease depending on the extent of within-cluster correlation.
- The regression coefficient size (β) remains unchanged, as clustering affects only standard errors.
- The significance of coefficients may decrease, as clustering typically inflates standard errors in the presence of high within-cluster correlation, making it harder to reject the null hypothesis.

Thus, clustering leads to more reliable inference, as it provides standard errors that correctly reflect the structure of the data.

Question

Consider two processes y_t and x_t where we have $t = 1, 2, \dots, T$ observations on each. Suppose that y_t represents a 10-year interest rate and x_t represents a three-month interest rate. Suppose we are interested in eventually forecasting y_t with the help of both past values of y as well as past values of x . With this information, answer the following questions:

- (a) Write down an ARMA(1,1) which also includes a lagged value for x .
 - (b) How do you know how many lags to include for y ? How do you know how many lags to include for ε (the error term)?
 - (c) How do you know how many lags to include for x ?
 - (d) Write down the h -step ahead value for y (y_{t+h}) that you put down in part (a).
 - (e) For $h = 1$, construct the forecast value of y ($\hat{y}_{t+h|t}$).
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Answer

(a) ARMA(1,1) with lagged value of x

The ARMA(1,1) model with a lagged value for x_t is given by:

$$y_t = \alpha + \phi_1 y_{t-1} + \theta_1 \varepsilon_{t-1} + \beta x_{t-1} + \varepsilon_t,$$

where:

- α : Constant term,
- ϕ_1 : Autoregressive coefficient for y_{t-1} ,
- θ_1 : Moving average coefficient for ε_{t-1} ,
- β : Coefficient for the lagged value of x_t ,
- ε_t : Error term.

(b) Determining the number of lags for y_t and ε_t

The number of lags for y_t and ε_t can be determined using:

- **Autocorrelation Function (ACF)**: To assess the dependence of y_t on its past values.
- **Partial Autocorrelation Function (PACF)**: To determine the appropriate order of the autoregressive process.
- Use statistical criteria like Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC) to choose the optimal number of lags.

(c) Determining the number of lags for x_t

The number of lags for x_t can be selected based on its impact on y_t :

- Conduct Granger causality tests to assess whether x_t helps predict y_t .
- Use ACF and PACF to examine the influence of x_t 's past values.
- Consider economic theory and prior literature to include relevant lags.

(d) h -step ahead forecast for y

The h -step ahead forecast for y_{t+h} is:

$$y_{t+h} = \alpha + \phi_1 y_{t+h-1} + \theta_1 \varepsilon_{t+h-1} + \beta x_{t+h-1}.$$

(e) Forecast for $h = 1$

For $h = 1$, the one-step ahead forecast of y_t is:

$$\hat{y}_{t+1|t} = \alpha + \phi_1 y_t + \theta_1 \varepsilon_t + \beta x_t,$$

where ε_t is typically replaced by its estimated value from the model.

Question

Consider two processes y_t and x_t where we have $t = 1, 2, \dots, T$ observations on each. Suppose that y_t represents a 10-year interest rate and x_t represents a three-month interest rate. Suppose we are interested in eventually forecasting y_t with the help of both past values of y as well as past values of x . With this information, answer the following questions:

- (a) Write down an ARMA(2,2) which also includes a lagged value for x .
 - (b) How do you know how many lags to include for y ? How do you know how many lags to include for ε (the error term)?
 - (c) How do you know how many lags to include for x ?
 - (d) Write down the h -step ahead value for y (y_{t+h}) that you put down in part (a).
 - (e) For $h = 1$, construct the forecast value of y ($\hat{y}_{t+h|t}$).
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Answer

(a) ARMA(2,2) with lagged value of x

The ARMA(2,2) model with a lagged value for x_t is:

$$y_t = \alpha + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \beta x_{t-1} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t,$$

where:

- α : Constant term,
- ϕ_1, ϕ_2 : Autoregressive coefficients,
- β : Coefficient for the lagged value of x_t ,
- θ_1, θ_2 : Moving average coefficients,
- ε_t : Error term.

(b) Determining the number of lags for y_t and ε_t

The number of lags for y_t and ε_t can be determined by:

- Examining the **Autocorrelation Function (ACF)** for significant correlations at specific lags.
- Examining the **Partial Autocorrelation Function (PACF)** to identify the order of the autoregressive process.
- Using model selection criteria such as Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC) to optimize the number of lags.

(c) Determining the number of lags for x_t

The number of lags for x_t can be determined by:

- Conducting Granger causality tests to assess the predictive value of x_t on y_t .
- Testing the significance of $x_{t-1}, x_{t-2}, \dots$ in the regression model using statistical inference.
- Considering economic theory to justify the inclusion of certain lags.

(d) h -step ahead forecast for y

The h -step ahead forecast for y_{t+h} based on the model in part (a) is:

$$y_{t+h} = \alpha + \phi_1 y_{t+h-1} + \phi_2 y_{t+h-2} + \beta x_{t+h-1} + \theta_1 \varepsilon_{t+h-1} + \theta_2 \varepsilon_{t+h-2}.$$

(e) Forecast for $h = 1$

For $h = 1$, the one-step ahead forecast of y_t is:

$$\hat{y}_{t+1|t} = \alpha + \phi_1 y_t + \phi_2 y_{t-1} + \beta x_t + \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1},$$

where ε_t is replaced by its estimated value from the model.

Question

You are given the regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

You want to test the null hypothesis:

$$H_0 : \beta_1^2 \beta_3 = \sqrt{\beta_2}$$

$$H_A : \beta_1^2 \beta_3 \neq \sqrt{\beta_2}$$

Show how you would construct the test statistic.

Answer

1. **Restate the Null Hypothesis**:

$$g(\beta_1, \beta_2, \beta_3) = \beta_1^2 \beta_3 - \sqrt{\beta_2}.$$

2. **Linearize g **: Use the first-order Taylor expansion around $(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)$:

$$g(\beta_1, \beta_2, \beta_3) \approx g(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3) + \frac{\partial g}{\partial \beta_1}(\beta_1 - \hat{\beta}_1) + \frac{\partial g}{\partial \beta_2}(\beta_2 - \hat{\beta}_2) + \frac{\partial g}{\partial \beta_3}(\beta_3 - \hat{\beta}_3),$$

where:

$$\frac{\partial g}{\partial \beta_1} = 2\hat{\beta}_1\hat{\beta}_3, \quad \frac{\partial g}{\partial \beta_2} = -\frac{1}{2\sqrt{\hat{\beta}_2}}, \quad \frac{\partial g}{\partial \beta_3} = \hat{\beta}_1^2.$$

3. **Test Statistic**: The test statistic is:

$$T = \frac{g(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)}{\sqrt{\text{Var}(g(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3))}},$$

where:

$$\text{Var}(g) = \begin{bmatrix} \frac{\partial g}{\partial \beta_1} & \frac{\partial g}{\partial \beta_2} & \frac{\partial g}{\partial \beta_3} \end{bmatrix} \text{Var}(\hat{\beta}) \begin{bmatrix} \frac{\partial g}{\partial \beta_1} \\ \frac{\partial g}{\partial \beta_2} \\ \frac{\partial g}{\partial \beta_3} \end{bmatrix}.$$

4. **Decision Rule**: Reject H_0 if $|T| > z_{\alpha/2}$, where $z_{\alpha/2}$ is the critical value from the standard normal distribution.
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Question

You are given the regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

You want to test the null hypothesis:

$$H_0 : \beta_1 \beta_3 = \beta_2^2$$

$$H_A : \beta_1 \beta_3 \neq \beta_2^2$$

Show how you would construct the test statistic.

Answer

1. **Restate the Null Hypothesis**:

$$g(\beta_1, \beta_2, \beta_3) = \beta_1\beta_3 - \beta_2^2$$

2. **Linearize g **: Using a first-order Taylor expansion:

$$g(\beta_1, \beta_2, \beta_3) \approx g(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3) + \frac{\partial g}{\partial \beta_1}(\beta_1 - \hat{\beta}_1) + \frac{\partial g}{\partial \beta_2}(\beta_2 - \hat{\beta}_2) + \frac{\partial g}{\partial \beta_3}(\beta_3 - \hat{\beta}_3),$$

where:

$$\frac{\partial g}{\partial \beta_1} = \hat{\beta}_3, \quad \frac{\partial g}{\partial \beta_2} = -2\hat{\beta}_2, \quad \frac{\partial g}{\partial \beta_3} = \hat{\beta}_1$$

3. **Test Statistic**:

$$T = \frac{g(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)}{\sqrt{\text{Var}(g(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3))}},$$

where:

$$\text{Var}(g) = \begin{bmatrix} \frac{\partial g}{\partial \beta_1} & \frac{\partial g}{\partial \beta_2} & \frac{\partial g}{\partial \beta_3} \end{bmatrix} \text{Var}(\hat{\beta}) \begin{bmatrix} \frac{\partial g}{\partial \beta_1} \\ \frac{\partial g}{\partial \beta_2} \\ \frac{\partial g}{\partial \beta_3} \end{bmatrix}.$$

4. **Decision Rule**: Reject H_0 if $|T| > z_{\alpha/2}$, where $z_{\alpha/2}$ is the critical value.

Question

5. In 2019, City X and City Y were two neighboring cities with similar economies. In 2020, City X implemented a significant increase in the minimum wage, while City Y did not make any changes to its minimum wage policy. Researchers collected employment data from a random sample of restaurants in both cities in the years before and after the minimum wage change to examine its effects on restaurant employment.

- (a) Using a difference-in-differences (DiD) identification strategy, formulate a linear probability model (LPM) to investigate whether the minimum wage increase in City X had an impact on the probability of restaurant employees being employed in 2021. Which coefficient in your model quantifies the effect of the minimum wage change?

- (b) Interpretation of the Coefficient of Interest: How would you interpret the coefficient associated with the minimum wage change in your LPM?
 - (c) Logit Model: Explain how the interpretation of the coefficient related to the minimum wage change would differ if you used a Logit model instead of an LPM.
 - (d) Parallel Trend Assumption: Define the Parallel Trend Assumption in the context of this study. What strategies or methods could you use to assess whether this assumption is satisfied using the data from the restaurants in City X and City Y?
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Answer

- (a) The DiD LPM can be specified as:

$$Y_{ict} = \alpha + \beta_1 D_{city} + \beta_2 D_{time} + \delta(D_{city} \cdot D_{time}) + \epsilon_{ict},$$

where:

- Y_{ict} : Binary indicator for employment of restaurant employees.
- D_{city} : Dummy for City X (= 1 if City X, = 0 if City Y).
- D_{time} : Dummy for post-2020 period (= 1 if year is 2021, = 0 otherwise).
- $D_{city} \cdot D_{time}$: Interaction term to capture the effect of the minimum wage increase.

The coefficient δ quantifies the effect of the minimum wage change on restaurant employment in City X.

- (b) The coefficient δ represents the change in the probability of restaurant employees being employed in City X after the minimum wage increase, relative to City Y.
- (c) In a Logit model, the interpretation of the coefficient related to the minimum wage change would reflect the log-odds of being employed rather than the direct change in probability. Marginal effects would need to be computed to quantify the impact on probabilities.
- (d) The Parallel Trend Assumption implies that in the absence of the minimum wage change, the trends in employment for City X and City Y would have been the same. To test this assumption, you could:

- Compare pre-2020 trends in employment for both cities.
 - Conduct a placebo test by applying the DiD framework to pre-treatment years only.
 - Include control variables to ensure similar baseline characteristics between cities.
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Question

8. Suppose we have data for U.S. interest rate data over the period 1960:Q1 to 2008:Q1 and we are interested in estimating a quarterly model of spread between a long-term and a short-term interest rate. Specifically, the interest rate spread (s_t) can be formed as the difference between the interest rate on a 10-year U.S. government bond ($r10_t$) and the rate on a three-month treasury bill ($Tbill_t$) as:

$$s_t = r10_t - Tbill_t$$

Suppose the interest rate spread in the fourth quarter of 2007 is 0.75, the interest rate spread in the first quarter of 2008 is 1.51 and the interest rate spread in the second quarter of 2008 is 1.61. Use the EViews output below to answer the following questions:

Dependent Variable: S Method: Least Squares Date: 12/30/09 Time: 10:28 Sample (adjusted): 1960Q3 2008Q1 Included observations: 191 after adjustments Convergence achieved after 3 iterations				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.387620	0.285227	4.864963	0.0000
AR(1)	1.109152	0.070829	15.65953	0.0000
AR(2)	-0.245386	0.070792	-3.466283	0.0007
R-squared	0.805374	Mean dependent var		1.378220
Adjusted R-squared	0.803304	S.D. dependent var		1.210806
S.E. of regression	0.536998	Akaike info criterion		1.609936
Sum squared resid	54.21291	Schwarz criterion		1.661019
Log likelihood	-150.7489	F-statistic		388.9784
Durbin-Watson stat	1.960866	Prob(F-statistic)		0.000000
Inverted AR Roots	.80	.31		

Figure 3: Comp Eviews Output

- Write the estimated equation given in the EViews output.
- Obtain the one-step ahead forecast.
- Obtain the one-step ahead forecast error.
- Obtain the two-step ahead forecast.
- Obtain the three-step ahead forecast.

Answer

- The estimated AR(2) equation from the EViews output is:

$$s_t = 1.387620 + 1.109152s_{t-1} - 0.245386s_{t-2} + \varepsilon_t$$

where:

- s_t : Interest rate spread at time t ,
- s_{t-1} : Interest rate spread at time $t - 1$,
- s_{t-2} : Interest rate spread at time $t - 2$,
- ε_t : Error term.

(b) The one-step ahead forecast ($t = Q1$ 2008) is:

$$\hat{s}_{2008:Q1} = 1.387620 + 1.109152(0.75) - 0.245386(1.51)$$

$$\hat{s}_{2008:Q1} = 1.387620 + 0.831864 - 0.370537 = 1.848947$$

(c) The one-step ahead forecast error ($e_{2008:Q1}$) is:

$$e_{2008:Q1} = s_{2008:Q1} - \hat{s}_{2008:Q1}$$

$$e_{2008:Q1} = 1.51 - 1.848947 = -0.338947$$

(d) The two-step ahead forecast ($t = Q2$ 2008) is:

$$\hat{s}_{2008:Q2} = 1.387620 + 1.109152(1.51) - 0.245386(0.75)$$

$$\hat{s}_{2008:Q2} = 1.387620 + 1.673827 - 0.184040 = 2.877407$$

(e) The three-step ahead forecast ($t = Q3$ 2008) is:

$$\hat{s}_{2008:Q3} = 1.387620 + 1.109152(1.61) - 0.245386(1.51)$$

$$\hat{s}_{2008:Q3} = 1.387620 + 1.785738 - 0.370537 = 2.802821$$

Explanation of Values in the Problem

The values used in the calculations for Parts (b), (c), (d), and (e) are provided in the problem statement. Below is a breakdown of their sources:

Part (b): One-step ahead forecast ($t = Q1$ 2008)

- **Where did 0.75 come from?**

This is the *interest rate spread* (s_{t-1}) for the previous quarter, i.e., Q4 2007, given in the problem:

"The interest rate spread in the fourth quarter of 2007 is 0.75."

- **Where did 1.51 come from?**

This is the *interest rate spread* (s_t) for Q1 2008, also explicitly provided in the problem:

"The interest rate spread in the first quarter of 2008 is 1.51."

Part (c): One-step ahead forecast error ($e_{2008:Q1}$)

- **Where did 1.51 come from?**

This is the *actual value* of s_t for Q1 2008, as stated in the problem:

"The interest rate spread in the first quarter of 2008 is 1.51."

- **Where did 1.848947 come from?**

This is the *predicted value* of $\hat{s}_t$ for Q1 2008, which was calculated in Part (b) using the AR(2) model:

$$\hat{s}_{2008:Q1} = 1.387620 + 1.109152(0.75) - 0.245386(1.51) = 1.848947$$

Part (d): Two-step ahead forecast ($t = Q2$ 2008)

- **Where did 1.51 come from?**

This is the *interest rate spread* (s_{t-1}) for Q1 2008, as provided in the problem:

"The interest rate spread in the first quarter of 2008 is 1.51."

- **Where did 0.75 come from?**

This is the *interest rate spread* (s_{t-2}) for Q4 2007, as stated in the problem:

"The interest rate spread in the fourth quarter of 2007 is 0.75."

Part (e): Three-step ahead forecast ($t = Q3$ 2008)

- **Where did 1.61 come from?**

This is the *interest rate spread* (s_{t-1}) for Q2 2008, as explicitly stated in the problem:

"The interest rate spread in the second quarter of 2008 is 1.61."

- **Where did 1.51 come from?**

This is the *interest rate spread* (s_{t-2}) for Q1 2008, as provided in the problem:

"The interest rate spread in the first quarter of 2008 is 1.51."

Summary Table of Values

Variable	Value	Source
s_{t-2} (Q4 2007)	0.75	Problem statement
s_{t-1} (Q1 2008)	1.51	Problem statement
s_t (Q2 2008)	1.61	Problem statement

[illegible]